The scaffold is the product.

On real, existing-repository bug-fixing (SWE-bench Lite, django/sympy-heavy), ExecuteSpec's agentic pipeline resolves **59%** (Anthropic) / **52%** (OpenAI) of issues — **above** the published Lite state-of-the-art (~48%). The **same models alone** — even handed the right file, even given five attempts — resolve just 3–10%. Our system delivers a **5–17× lift**. That gap is the reason to exist, and it is now measured on the official scorer.

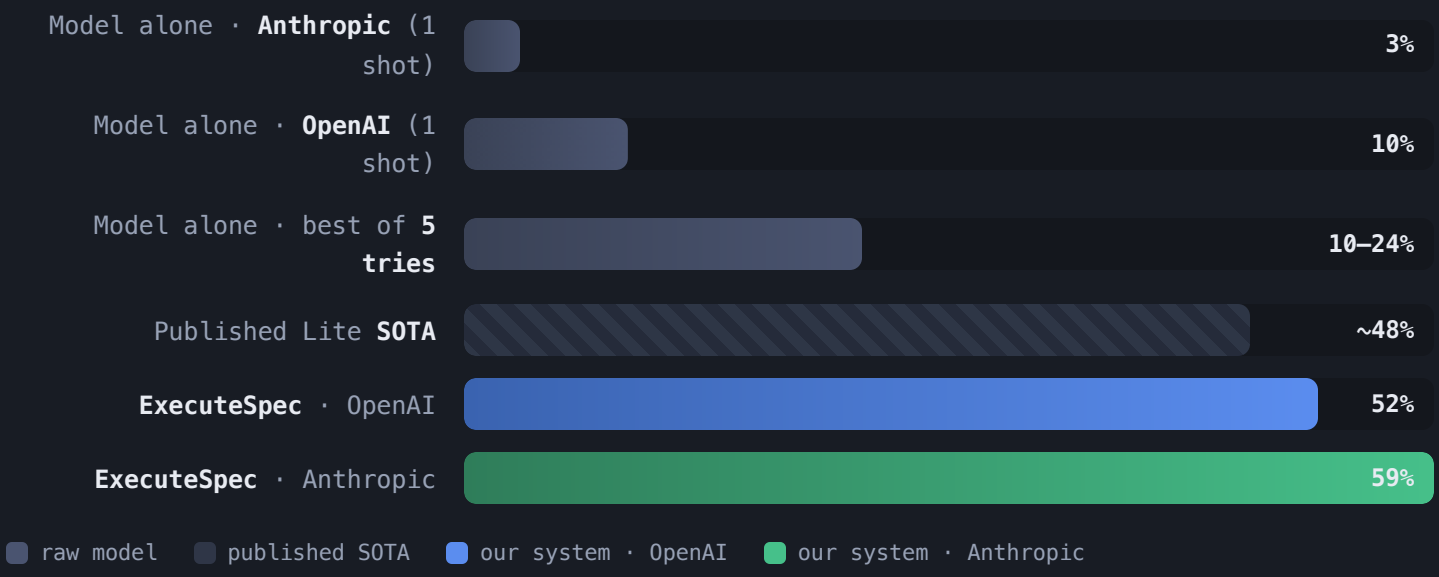
59%
RESOLVED · ANTHROPIC
17 / 29 · vs ~48% Lite SOTA

52%
RESOLVED · OPENAI
15 / 29

5–17×
LIFT OVER MODEL ALONE
+41 to +55 points

88%
LOCALIZATION HIT-RATE
lands the exact right file

THE LIFT — MODEL ALONE VS. THE SYSTEM



Even with **five samples**, the raw model tops out at 10–24%. The lever is not sampling — it is the **agentic loop**: investigate → reproduce the bug → iterate → verify against the repo's own tests. That is the defensible asset.

THE DIFFERENTIATOR — EXISTING-REPO WORK, WHERE THE LIFT IS LARGEST

Competitors largely chase **greenfield** generation, where a strong base model already does well and the moat is thin. ExecuteSpec's edge is **existing-repo PATCH** — landing a correct change in an unfamiliar, large codebase — which is exactly where the raw model collapses (3–10%) and our scaffold multiplies it 5–17×. As model prices rise, a scaffold that extracts this much more value per token compounds in importance.

DEPTH — WHERE THE WINS LAND (ANTHROPIC, OFFICIAL SCORER)

COMPLEXITY BAND	RESOLVED	RATE
Low	7 / 10	70%
Medium	6 / 11	55%
High	4 / 8	50%
Total	**17 / 29**	59%

REPOSITORY	RESOLVED
django	9 / 13
sympy	4 / 7
scikit-learn	2 / 2
matplotlib	1 / 3
pytest	1 / 1

PER-INSTANCE RESULTS — THE FULL STRATIFIED SAMPLE (AUDITABLE)

Every test case in the 30-instance stratified sample (LOW 10 · MED 12 · HIGH 8), with its **official-harness** System (Anthropic) outcome — so the 59% headline is auditable, not just asserted. The headline **17 / 29** reflects 29 scored (one MEDIUM instance was excluded at scoring); the **12 misses** below are the authoritative improvement target.

TEST CASE (SWE-BENCH INSTANCE ID)	REPO	BAND	SYSTEM · ANTHROPIC
django__django-15347	django	LOW	✓ resolved
sympy__sympy-17139	sympy	LOW	✓ resolved
django__django-13964	django	LOW	✓ resolved
sympy__sympy-14817	sympy	LOW	✓ resolved
django__django-14580	django	LOW	✓ resolved
sympy__sympy-15609	sympy	LOW	✓ resolved
django__django-13028	django	LOW	✓ resolved
django__django-15738	django	LOW	✗ failed
sympy__sympy-24102	sympy	LOW	✗ failed

TEST CASE (SWE-BENCH INSTANCE ID)	REPO	BAND	SYSTEM · ANTHROPIC
django__django-14534	django	LOW	✗ failed
scikit-learn__scikit-learn-12471	scikit-learn	MED	✓ resolved
django__django-13158	django	MED	✓ resolved
django__django-11964	django	MED	✓ resolved
sympy__sympy-14396	sympy	MED	✓ resolved
django__django-13590	django	MED	✓ resolved
django__django-14855	django	MED	✓ resolved
django__django-11422	django	MED	✓ resolved
sphinx-doc__sphinx-10325	sphinx	MED	✗ failed
django__django-12589	django	MED	✗ failed
pydata__xarray-3364	xarray	MED	✗ failed
sphinx-doc__sphinx-8435	sphinx	MED	✗ failed
matplotlib__matplotlib-22835	matplotlib	MED	✗ failed
scikit-learn__scikit-learn-10297	scikit-learn	HIGH	✓ resolved
matplotlib__matplotlib-23913	matplotlib	HIGH	✓ resolved
django__django-13768	django	HIGH	✓ resolved
pytest-dev__pytest-7373	pytest	HIGH	✓ resolved
matplotlib__matplotlib-18869	matplotlib	HIGH	✗ failed
django__django-11019	django	HIGH	✗ failed
sympy__sympy-16106	sympy	HIGH	✗ failed
sympy__sympy-14317	sympy	HIGH	✗ failed

Raw model output — same 29-instance sample. What each model resolves *alone* (single LLM call, oracle file, official-harness scored) vs. inside ExecuteSpec — the contrast the per-instance table above makes case-by-case for the system.

PROVIDER	RAW MODEL · 1 ATTEMPT	RAW MODEL · BEST OF 5	EXECUTESPEC (SYSTEM)
Anthropic · claude-sonnet-4-6	2 / 29 · 7%	3 / 29 · 10%	17 / 29 · 59%
OpenAI · gpt-5.4	3 / 29 · 10%	7 / 29 · 24%	15 / 29 · 52%

Provenance. System results are per-instance from the official SWE-bench harness (the table above). The raw-model baseline is the verified **aggregate** — per-instance raw-model runs were not retained, so a per-row raw-model column would require re-running the baseline.

ECONOMICS

ANTHROPIC · ACCURACY-FIRST

$0.71 / issue

~8 min / issue · highest resolve rate

OPENAI · THROUGHPUT-FIRST

$0.35 / issue

~2 min / issue · ~2× cheaper, ~4× faster

METHOD & RIGOR

- **Official scorer.** All numbers are from the **official SWE-bench harness** (v4.1.0), which runs each repository's real test suite on the saved patch — not a proxy. Re-scoring costs $0 in model spend.
- **Blind, not oracle.** The agent sees only the issue text; the hidden acceptance tests are applied externally at scoring. No test leakage into the prompt.

Scope & caveats. N = 29 (one stratified sample across Low/Medium/High); directional, not a leaderboard submission. SWE-bench **Live** figures remain unverified pending the Live harness and are excluded here. Gemini excluded from this cut. Numbers are reproducible from saved patches at $0 model cost.

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